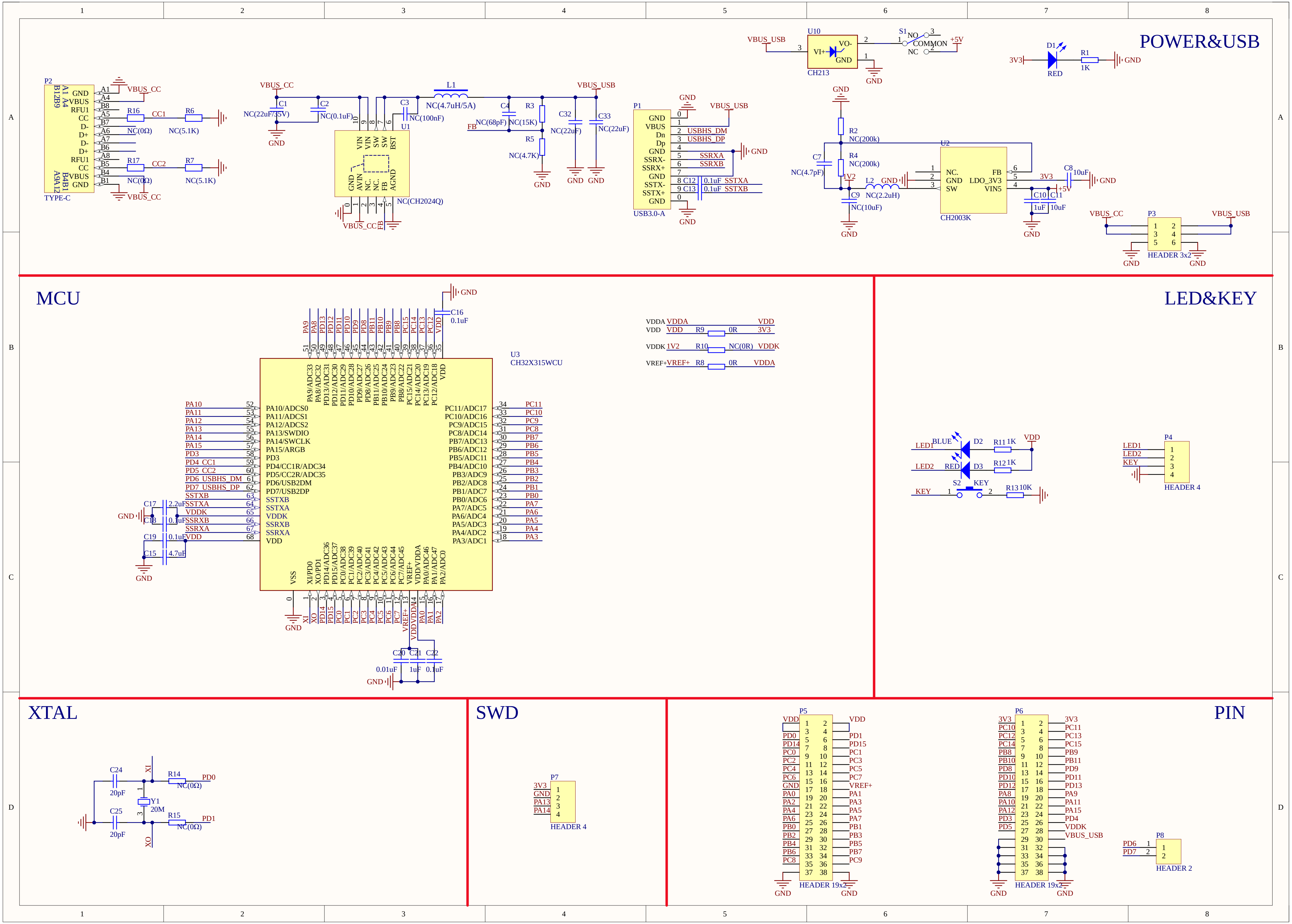
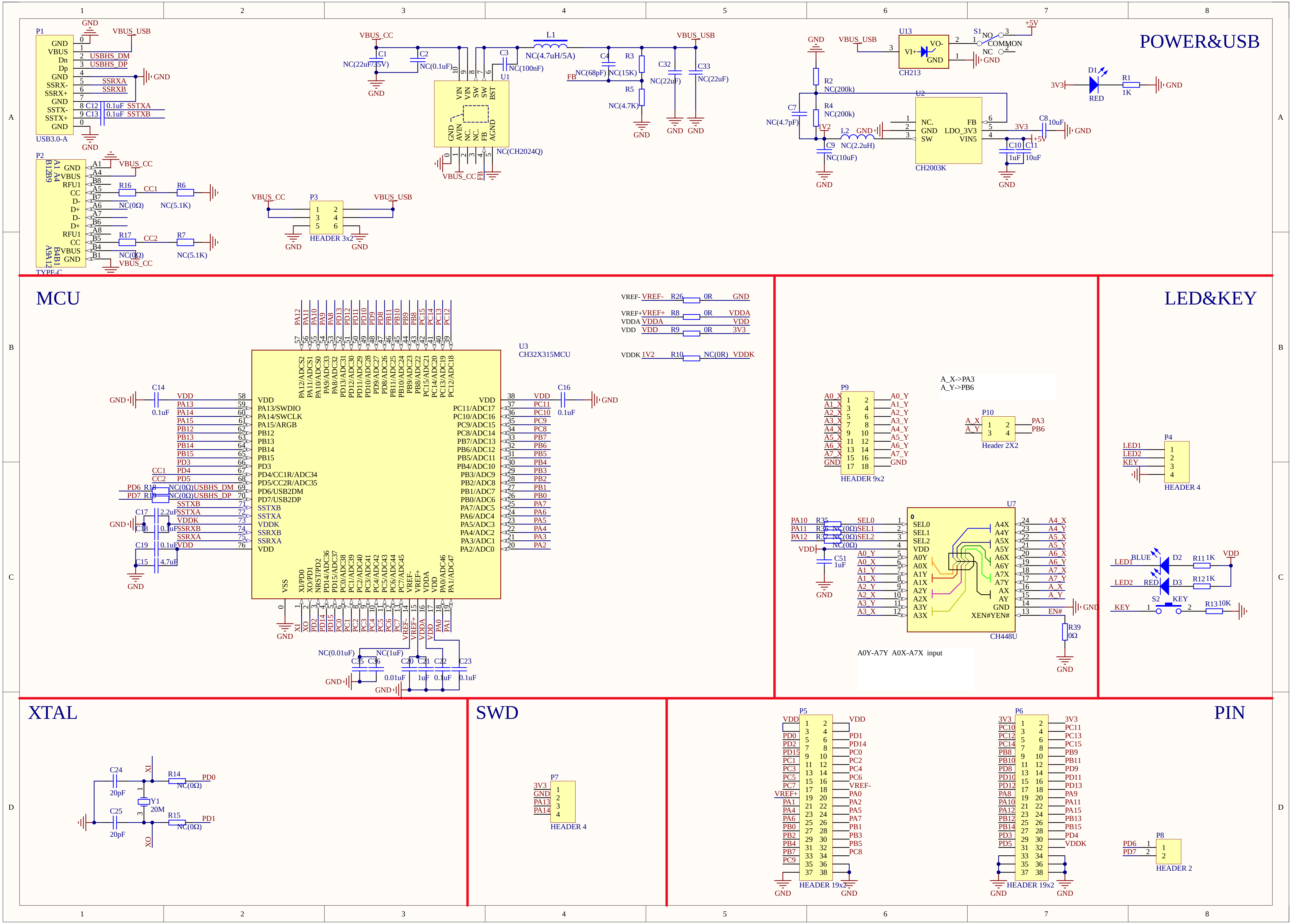
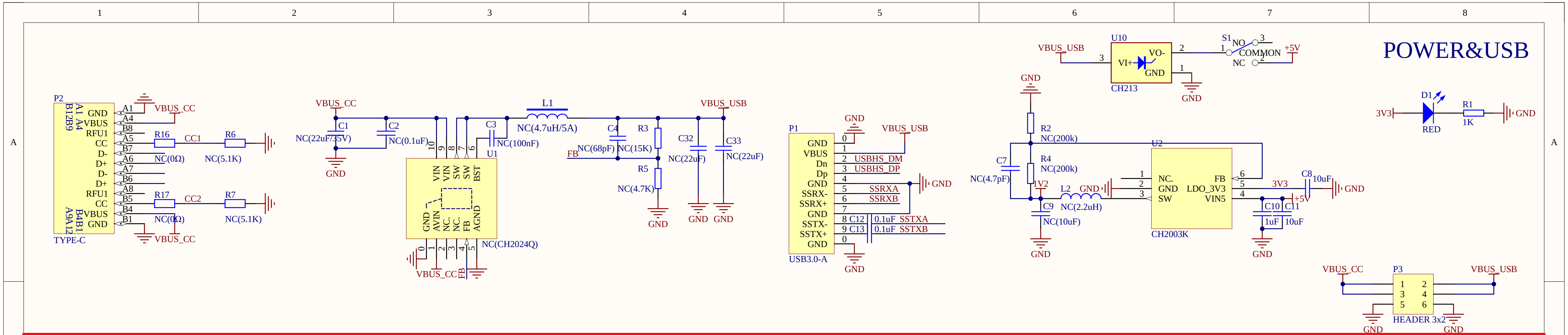
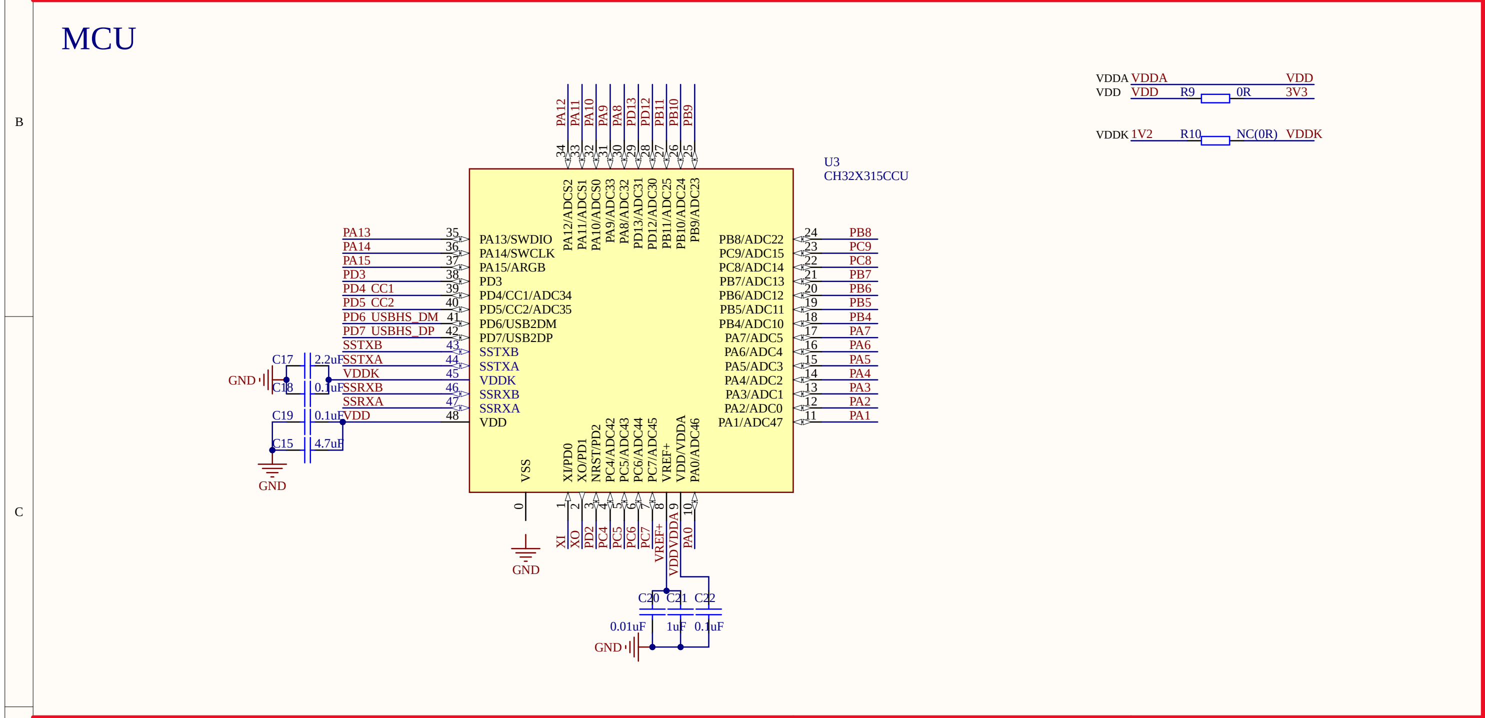

POWER&USB
MCU
LED&KEY
XTAL
SWD
PIN
1
2
3
4
5
6
7
8
A
B
C
D
P2
TYPE-C
GND
VBUS
RFU1
CC
D-
D+
VBUS_CC
R16
NC(0Ω)
CC1
R6
NC(5.1K)
R17
CC2
R7
C1
NC(22uF/35V)
C2
NC(0.1uF)
C3
NC(100nF)
U1
NC(CH2024Q)
VIN
SW
BST
AVIN
NC.
FB
AGND
L1
NC(4.7uH/5A)
C4
NC(68pF)
R3
NC(15K)
R5
NC(4.7K)
C32
NC(22uF)
C33
VBUS_USB
P1
USB3.0-A
SSRX-
SSRX+
SSTX-
SSTX+
USBHS_DM
USBHS_DP
SSRXA
SSRXB
C12
C13
0.1uF
SSTXA
SSTXB
U10
CH213
VI+
VO-
S1
NO
COMMON
NC
+5V
D1
RED
R1
1K
3V3
R2
NC(200k)
R4
C7
NC(4.7pF)
1V2
L2
NC(2.2uH)
C9
NC(10uF)
U2
CH2003K
LDO_3V3
VIN5
C8
10uF
C10
1uF
C11
P3
HEADER 3x2
U3
CH32X315WCU
PA10/ADCS0
PA11/ADCS1
PA12/ADCS2
PA13/SWDIO
PA14/SWCLK
PA15/ARGB
PD3
PD4/CC1R/ADC34
PD5/CC2R/ADC35
PD6/USB2DM
PD7/USB2DP
SSTXB
SSTXA
VDDK
SSRXB
SSRXA
VDD
PC11/ADC17
PC10/ADC16
PC9/ADC15
PC8/ADC14
PB7/ADC13
PB6/ADC12
PB5/ADC11
PB4/ADC10
PB3/ADC9
PB2/ADC8
PB1/ADC7
PB0/ADC6
PA7/ADC5
PA6/ADC4
PA5/ADC3
PA4/ADC2
PA3/ADC1
VSS
XI/PD0
XO/PD1
PD14/ADC36
PD15/ADC37
PC0/ADC38
PC1/ADC39
PC2/ADC40
PC3/ADC41
PC4/ADC42
PC5/ADC43
PC6/ADC44
PC7/ADC45
VREF+
VDD/VDDA
PA0/ADC46
PA1/ADC47
PA2/ADC0
PA9/ADC33
PA8/ADC32
PD13/ADC31
PD12/ADC30
PD11/ADC29
PD10/ADC28
PD9/ADC27
PD8/ADC26
PB11/ADC25
PB10/ADC24
PB9/ADC23
PB8/ADC22
PC15/ADC21
PC14/ADC20
PC13/ADC19
PC12/ADC18
C16
C17
2.2uF
C18
C19
C15
4.7uF
C20
0.01uF
C21
C22
VDDA
R9
0R
R10
NC(0R)
VREF+
R8
LED1
BLUE
D2
R11
LED2
D3
R12
S2
KEY
R13
10K
P4
HEADER 4
C24
20pF
C25
XI
XO
Y1
20M
R14
PD0
R15
PD1
P7
P5
HEADER 19x2
P6
P8
HEADER 2


POWER&USB
P1
USB3.0-A
GND
VBUS
Dn
Dp
SSRX-
SSRX+
SSTX-
SSTX+
VBUS_USB
USBHS_DM
USBHS_DP
SSRXA
SSRXB
C12 0.1uF SSTXA
C13 0.1uF SSTXB
P2
TYPE-C
VBUS_CC
CC1
CC2
RFU1
CC
D-
D+
R16 NC(0Ω)
R6 NC(5.1K)
R17 NC(0Ω)
R7 NC(5.1K)
P3
HEADER 3x2
VBUS_CC
C1 NC(22uF/35V)
C2 NC(0.1uF)
C3 NC(100nF)
U1
NC(CH2024Q)
VIN
SW
BST
AVIN
NC.
FB
AGND
L1 NC(4.7uH/5A)
C4 NC(68pF)
R3 NC(15K)
R5 NC(4.7K)
C32 NC(22uF)
C33 NC(22uF)
U13
CH213
VI+
VO-
S1
NO
COMMON
NC
+5V
D1
RED
R1 1K
3V3
R2 NC(200k)
R4 NC(200k)
C7 NC(4.7pF)
1V2
L2 NC(2.2uH)
C9 NC(10uF)
U2
CH2003K
NC.
GND
SW
FB
LDO_3V3
VIN5
C8 10uF
C10 1uF
C11 10uF
MCU
U3
CH32X315MCU
VREF- R26 0R GND
VREF+ R8 0R VDDA
VDDA VDD
VDD R9 0R 3V3
VDDK 1V2 R10 NC(0R) VDDK
C14 0.1uF
C16 0.1uF
PA13/SWDIO
PA14/SWCLK
PA15/ARGB
PB12
PB13
PB14
PB15
PD3
PD4/CC1R/ADC34
PD5/CC2R/ADC35
PD6/USB2DM
PD7/USB2DP
SSTXB
SSTXA
VDDK
SSRXB
SSRXA
R18 NC(0Ω) USBHS_DM
R19 NC(0Ω) USBHS_DP
C17 2.2uF
C18 0.1uF
C19 0.1uF
C15 4.7uF
PC11/ADC17
PC10/ADC16
PC9/ADC15
PC8/ADC14
PB7/ADC13
PB6/ADC12
PB5/ADC11
PB4/ADC10
PB3/ADC9
PB2/ADC8
PB1/ADC7
PB0/ADC6
PA7/ADC5
PA6/ADC4
PA5/ADC3
PA4/ADC2
PA3/ADC1
PA2/ADC0
PA12/ADCS2
PA11/ADCS1
PA10/ADCS0
PA9/ADC33
PA8/ADC32
PD13/ADC31
PD12/ADC30
PD11/ADC29
PD10/ADC28
PD9/ADC27
PD8/ADC26
PB11/ADC25
PB10/ADC24
PB9/ADC23
PB8/ADC22
PC15/ADC21
PC14/ADC20
PC13/ADC19
PC12/ADC18
VSS
XI/PD0
XO/PD1
NRST/PD2
PD14/ADC36
PD15/ADC37
PC0/ADC38
PC1/ADC39
PC2/ADC40
PC3/ADC41
PC4/ADC42
PC5/ADC43
PC6/ADC44
PC7/ADC45
VREF-
VREF+
VDDA
VDD
PA0/ADC46
PA1/ADC47
C35 NC(0.01uF)
C36 NC(1uF)
C20 0.01uF
C21 1uF
C22 0.1uF
C23 0.1uF
LED&KEY
A_X->PA3
A_Y->PB6
P9
HEADER 9x2
A0_X A1_X A2_X A3_X A4_X A5_X A6_X A7_X
A0_Y A1_Y A2_Y A3_Y A4_Y A5_Y A6_Y A7_Y
P10
Header 2X2
A_X A_Y PA3 PB6
P4
HEADER 4
LED1 LED2 KEY
U7
CH448U
SEL0 SEL1 SEL2
A0Y A0X A1Y A1X A2Y A2X A3Y A3X
A4X A4Y A5X A5Y A6X A6Y A7X A7Y
AX AY
XEN#YEN#
EN#
R35 R36 NC(0Ω) R37 NC(0Ω)
C51 1uF
R39 0Ω
A0Y-A7Y A0X-A7X input
D2 BLUE
R11 1K
D3 RED
R12 1K
S2 KEY
R13 10K
XTAL
C24 20pF
C25 20pF
Y1 20M
XI XO
R14 NC(0Ω) PD0
R15 NC(0Ω) PD1
SWD
P7
HEADER 4
3V3 GND PA13 PA14
PIN
P5
HEADER 19x2
VDD PD0 PD2 PD15 PC1 PC3 PC5 PC7 VREF+ PA1 PA4 PA6 PB0 PB2 PB4 PB7 PC9
VDD PD1 PD14 PC0 PC2 PC4 PC6 VREF- PA0 PA2 PA5 PA7 PB1 PB3 PB5 PC8
P6
HEADER 19x2
3V3 PC10 PC12 PC14 PB8 PB10 PD8 PD10 PD12 PA8 PA10 PA12 PB12 PB14 PD3 PD5
3V3 PC11 PC13 PC15 PB9 PB11 PD9 PD11 PD13 PA9 PA11 PA15 PB13 PB15 PD4 VDDK
P8
HEADER 2
PD6 PD7

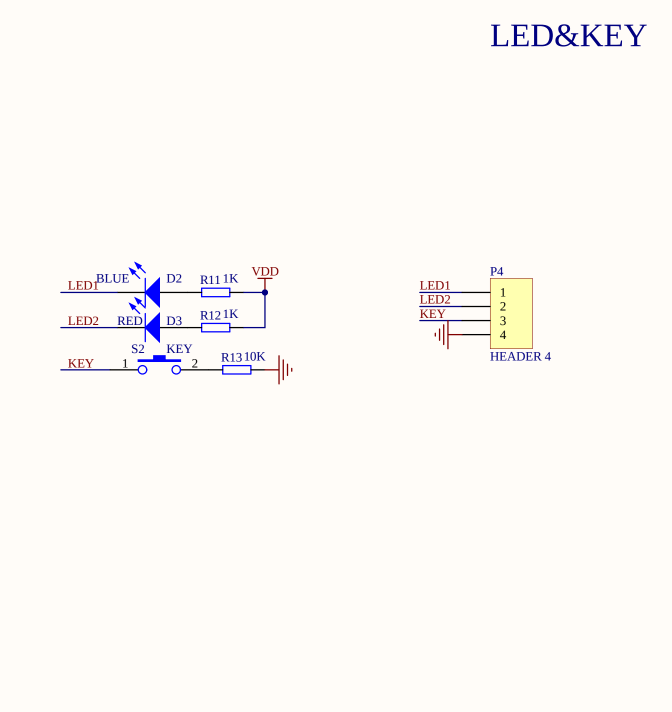
POWER&USB



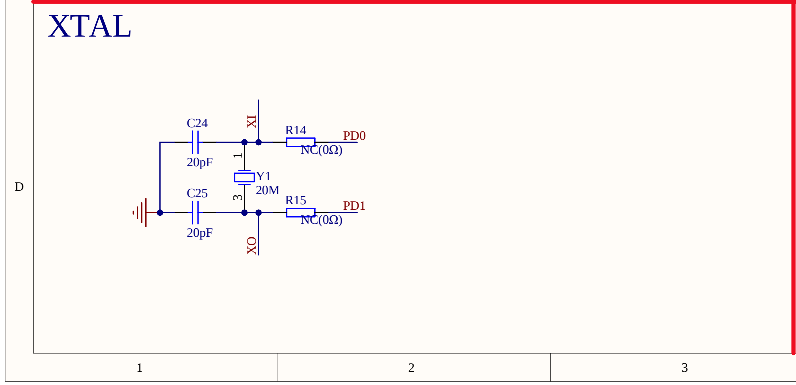
MCU



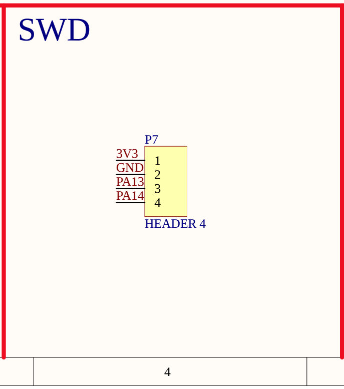
LED&KEY



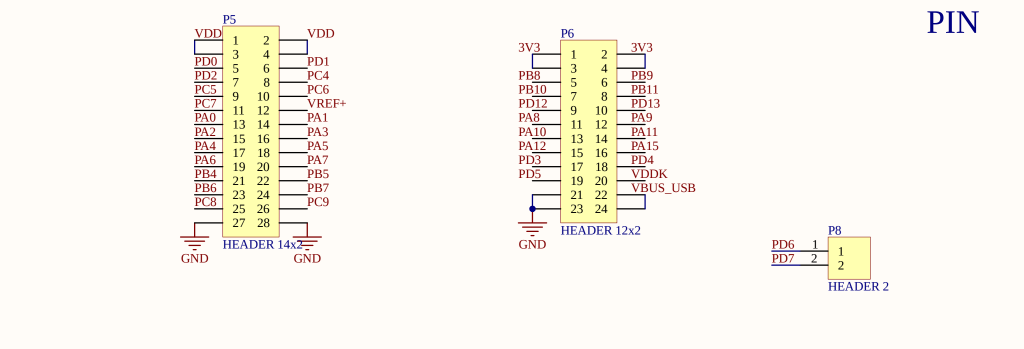
XTAL

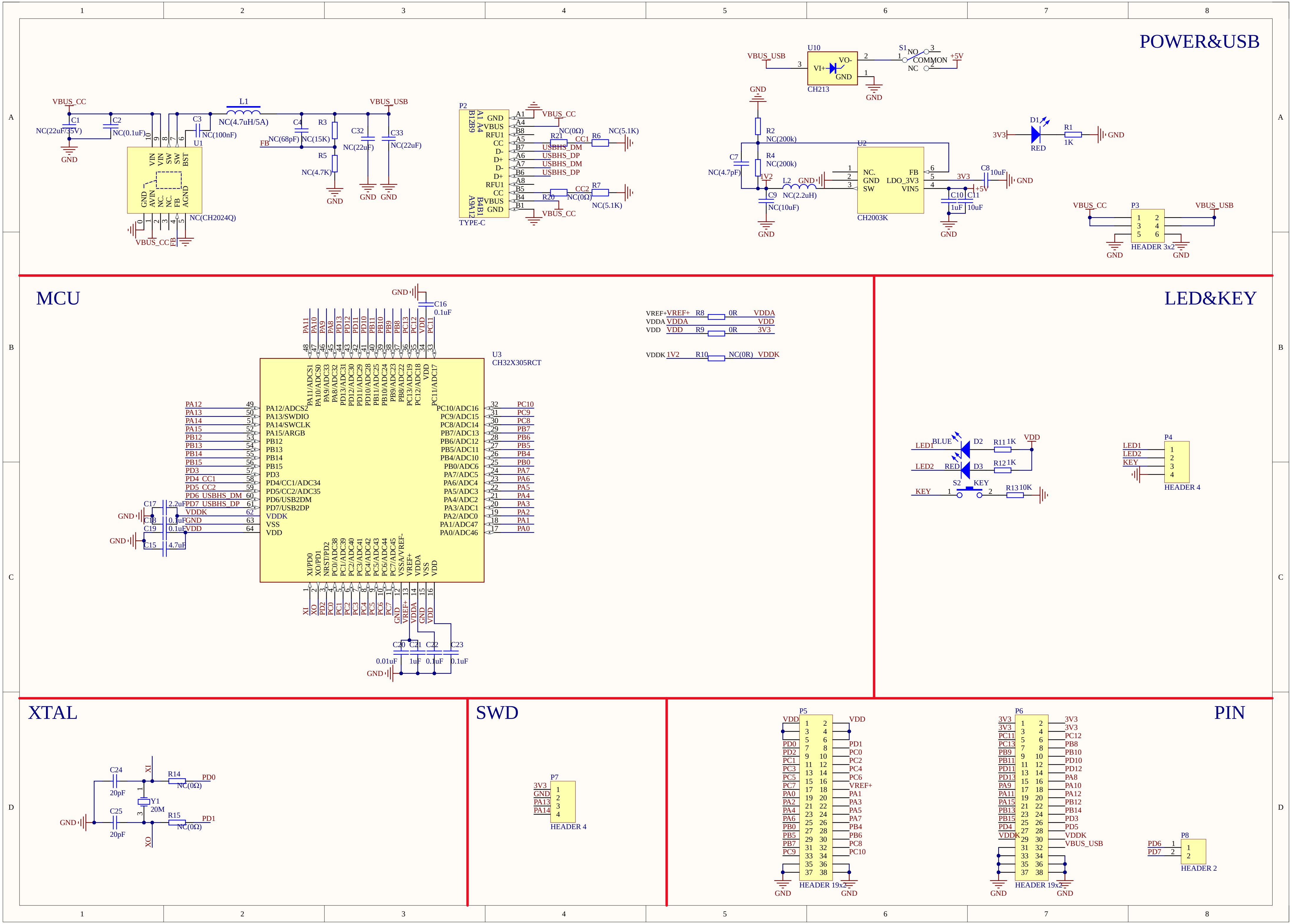


SWD



PIN




POWER&USB
VBUS_CC
C1
NC(22uF/35V)
C2
NC(0.1uF)
GND
C3
NC(100nF)
L1
NC(4.7uH/5A)
U1
VIN
VIN
SW
SW
BST
GND
AVIN
NC.
NC.
FB
AGND
NC(CH2024Q)
VBUS_CC
FB
C4
NC(68pF)
R3
NC(15K)
R5
NC(4.7K)
C32
NC(22uF)
C33
NC(22uF)
VBUS_USB
FB
GND
P2
TYPE-C
GND
VBUS
RFU1
CC
D-
D+
A1 A4
B12B9
B4B1
A9A12
VBUS_CC
R21
NC(0Ω)
CC1
R6
NC(5.1K)
USBHS_DM
USBHS_DP
USBHS_DM
USBHS_DP
CC2
R7
R20
NC(0Ω)
NC(5.1K)
VBUS_CC
VBUS_USB
U10
VI+
VO-
GND
CH213
S1
NO
COMMON
NC
+5V
GND
R2
NC(200k)
C7
NC(4.7pF)
R4
NC(200k)
1V2
L2
NC(2.2uH)
C9
NC(10uF)
U2
NC.
GND
SW
FB
LDO_3V3
VIN5
CH2003K
3V3
C8
10uF
+5V
C10
1uF
C11
10uF
3V3
D1
RED
R1
1K
GND
VBUS_CC
P3
VBUS_USB
HEADER 3x2
MCU
GND
C16
0.1uF
PA11 PA10 PA9 PA8 PD13 PD12 PD11 PD10 PB11 PB10 PB9 PB8 PC13 PC12 VDD PC11
U3
CH32X305RCT
PA11/ADCS1
PA10/ADCS0
PA9/ADC33
PA8/ADC32
PD13/ADC31
PD12/ADC30
PD11/ADC29
PD10/ADC28
PB11/ADC25
PB10/ADC24
PB9/ADC23
PB8/ADC22
PC13/ADC19
PC12/ADC18
VDD
PC11/ADC17
PA12/ADCS2
PA13/SWDIO
PA14/SWCLK
PA15/ARGB
PB12
PB13
PB14
PB15
PD3
PD4/CC1/ADC34
PD5/CC2/ADC35
PD6/USB2DM
PD7/USB2DP
VDDK
VSS
VDD
PC10/ADC16
PC9/ADC15
PC8/ADC14
PB7/ADC13
PB6/ADC12
PB5/ADC11
PB4/ADC10
PB0/ADC6
PA7/ADC5
PA6/ADC4
PA5/ADC3
PA4/ADC2
PA3/ADC1
PA2/ADC0
PA1/ADC47
PA0/ADC46
XI/PD0
XO/PD1
NRST/PD2
PC0/ADC38
PC1/ADC39
PC2/ADC40
PC3/ADC41
PC4/ADC42
PC5/ADC43
PC6/ADC44
PC7/ADC45
VSSA/VREF-
VREF+
VDDA
VSS
VDD
PA12 PA13 PA14 PA15 PB12 PB13 PB14 PB15 PD3
PD4_CC1
PD5_CC2
PD6_USBHS_DM
PD7_USBHS_DP
VDDK
GND
VDD
PC10 PC9 PC8 PB7 PB6 PB5 PB4 PB0 PA7 PA6 PA5 PA4 PA3 PA2 PA1 PA0
C17
2.2uF
C18
0.1uF
C19
0.1uF
C15
4.7uF
GND
XI XO PD2 PC0 PC1 PC2 PC3 PC4 PC5 PC6 PC7 GND VREF+ VDDA GND VDD
C20
0.01uF
C21
1uF
C22
0.1uF
C23
0.1uF
GND
VREF+ VREF+ R8 0R VDDA
VDDA VDDA VDD
VDD VDD R9 0R 3V3
VDDK 1V2 R10 NC(0R) VDDK
LED&KEY
LED1
BLUE
D2
R11 1K
VDD
LED2
RED
D3
R12 1K
KEY
S2
KEY
R13 10K
LED1
LED2
KEY
P4
HEADER 4
XTAL
C24
20pF
XI
R14
NC(0Ω)
PD0
Y1
20M
GND
C25
20pF
R15
NC(0Ω)
PD1
XO
SWD
P7
3V3
GND
PA13
PA14
HEADER 4
PIN
P5
VDD
VDD
PD0 PD2 PC1 PC3 PC5 PC7 PA0 PA2 PA4 PA6 PB0 PB5 PB7 PC9
PD1 PC0 PC2 PC4 PC6 VREF+ PA1 PA3 PA5 PA7 PB4 PB6 PC8 PC10
GND
GND
HEADER 19x2
P6
3V3 3V3 PC11 PC13 PB9 PB11 PD11 PD13 PA9 PA11 PA15 PB13 PB15 PD4 VDDK
3V3 3V3 PC12 PB8 PB10 PD10 PD12 PA8 PA10 PA12 PB12 PB14 PD3 PD5 VDDK VBUS_USB
GND
GND
HEADER 19x2
P8
PD6
PD7
HEADER 2
1 2 3 4 5 6 7 8
A B C D